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Centromere and telomere sequence alterations reflect the rapid genome evolution within the carnivorous plant genus *Genlisea*

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SUMMARY

Linear chromosomes of eukaryotic organisms invariably possess centromeres and telomeres to ensure proper chromosome segregation during nuclear divisions and to protect the chromosome ends from deterioration and fusion, respectively. While centromeric sequences may differ between species, with arrays of tandemly repeated sequences and retrotransposons being the most abundant sequence types in plant centromeres, telomeric sequences are usually highly conserved among plants and other organisms. The genome size of the carnivorous genus *Genlisea* (Lentibulariaceae) is highly variable. Here we study evolutionary sequence plasticity of these chromosomal domains at an intrageneric level. We show that *Genlisea nigrocaulis* (1C = 86 Mbp; 2n = 40) and *G. hispidula* (1C = 1550 Mbp; 2n = 40) differ as to their DNA composition at centromeres and telomeres. *G. nigrocaulis* and its close relative *G. pygmaea* revealed mainly 161 bp tandem repeats, while *G. hispidula* and its close relative *G. subglabra* displayed a combination of four retroelements at centromeric positions. *G. nigrocaulis* and *G. pygmaea* chromosome ends are characterized by the Arabidopsis-type telomeric repeats (TTTAGGG); *G. hispidula* and *G. subglabra* instead revealed two intermingled sequence variants (TTCAGG and TTTCAGG). These differences in centromeric and, surprisingly, also in telomeric DNA sequences, uncovered between groups with on average a > 9-fold genome size difference, emphasize the fast genome evolution within this genus. Such intrageneric evolutionary alteration of telomeric repeats with cytosine in the guanine-rich strand, not yet known for plants, might impact the epigenetic telomere chromatin modification.

Keywords: Lentibulariaceae, *Genlisea nigrocaulis*, *G. hispidula*, centromeric tandem repeat, centromeric retrotransposons, plant telomeric repeat variants, telomerase, genome evolution.

INTRODUCTION

The nuclear genome of eukaryotic organisms is packed into chromosomes. Although the number and size of chromosomes may vary even between closely related species, linear eukaryotic chromosomes invariably possess as essential constituents a centromere and two telomeres. The centromere is required for sister chromatid cohesion until it becomes involved in chromosome segregation during nuclear divisions. Telomeres protect chromosome ends from degradation and fusion and prevent chromosome shortening during DNA replications.

Although centromere and telomere functions are highly conserved, their DNA sequences show different degrees of

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conservation across eukaryotic organisms. In vascular plants, the heptanucleotide repeat (TTTAGGG) which was first discovered in *Arabidopsis thaliana* (Richards and Ausubel, 1988) is the most common telomere sequence (Fuchs *et al.*, 1995). Nevertheless, this plant-type telomere repeat is replaced completely or partially by other repeats such as (TTAGGG) or (TTGGGG) or by unknown sequences within families of the monocotyledonous order Asparagales (Adams *et al.*, 2001; Weiss and Scherthan, 2002; Sykorova *et al.*, 2003c, 2006). Telomere repeat alteration was also detected within species of the dicotyledonous family Solanaceae (Sykorova *et al.*, 2003a; Peska *et al.*, 2015).

Variation of centromere-associated sequences is even more widespread. From numerous centromere DNA sequences reported for plants and animals, centromere-specific tandem repeats are the most common. A typical characteristic of these centromeric repeats is their rapid divergence (Henikoff *et al.*, 2001; Ma *et al.*, 2007). A comparative analysis of centromeric repeats in hundreds of plant and animal species revealed that the overall similarity of these elements is very low (Melters *et al.*, 2013). Variation in sequence and length of repeated monomers of centromeric repeats has been detected e.g. in *A. thaliana* (centromeric repeat family pAL1) (Martinez-Zapater *et al.*, 1986), *Oryza sativa* (CentO, formerly RCS2 family) (Dong *et al.*, 1998; Cheng *et al.*, 2002) and barley (AGGGAG) (Hudakova *et al.*, 2001). Other important components of centromeric DNA are transposable elements. Centromeres of a number of angiosperm species are colonized by centromeric retrotransposons (CR) belonging to the CRM (centromeric retrotransposon of maize) clade of chromoviruses, a lineage of *Ty3/gypsy* retrotransposons possessing an integrase chromodomain (Gorinsek *et al.*, 2004; Neumann *et al.*, 2011). Centromeric tandem repeats and CRs, either alone or together, contribute to the structure of centromeres which may span from hundreds of kbp to several Mbp in different species, and vary regarding their extension even among chromosomes of a complement (Hosouchi *et al.*, 2002; Jiang *et al.*, 2003; Plohl *et al.*, 2014). However only the core part of such extended and complex structures is defined as functional centromere which binds kinetochore proteins and contains the centromeric histone variant cenH3 (for review see Houben and Schubert, 2003; Jiang *et al.*, 2003; Fukagawa and Earnshaw, 2014). Further (peri)centromere-specific histone marks are histone H2A phosphorylated at threonine 133 in maize (Dong and Han, 2012) or at threonine 120/121 (see Materials and Methods and Demidov *et al.* (2014)) and histone H3 phosphorylated at serines 10 or 28 (Houben *et al.*, 1999) in all tested plants.

The carnivorous genus *Genlisea* comprises at least 29 species distributed in South and Central America and in Africa (Fleischmann, 2012) (Figure 1). This genus gained increasing scientific interest since *G. aurea* (63.6 Mbp/1C) and *G. margaretae* (63.4 Mbp/1C) were shown to possess the smallest nuclear genome sizes recorded for angiosperms (Greilhuber *et al.*, 2006). Later analyses including our own measurements could not confirm the ultra-small genome size for *G. margaretae* (Fleischmann, 2012; Veleba *et al.*, 2014). However, with *G. tuberosa* (approximately 63 Mbp/1C) another species with an ultra-small genome was reported (Fleischmann *et al.*, 2014). Other *Genlisea* species, such as *G. hispidula* (1417-1510 Mbp/1C) and *G. subglabra* (1471 Mbp/1C), were reported having an up to 24-fold larger genome (Greilhuber *et al.*, 2006; Veleba *et al.*, 2014). In spite of a very similar morphology, *Genlisea* species display with a 25-fold difference one of the largest genome size ranges reported within a genus.

The genomes of *G. nigrocaulis* and *G. hispidula* (86 Mbp and 1550 Mbp, respectively) were sequenced for a comprehensive comparative analysis and revealed divergent genome size evolution within this genus (Vu *et al.*, 2015). These two species have the same chromosome number (2n = 40), but, in correlation with their difference in genome size, they possess different proportions of repetitive DNA. Screening repetitive DNA profiles, we identified a 161-bp tandem repeat (GnCent satellite) for *G. nigrocaulis*, and four families of potentially centromeric retrotransposons (GhCR) for *G. hispidula* (Vu *et al.*, 2015).

Here we confirm the centromeric localization of the respective repeats by fluorescence *in situ* hybridization (FISH) in combination with immunofluorescence of centromere-specific histone marks in both species and their close relatives. Furthermore, studying the telomere composition of both species, we detected an alteration in telomere repeats within the genus. The Arabidopsis-type telomere repeat present in *G. nigrocaulis* were replaced by two variant sequences (TTCAGG) and (TTTCAGG) in *G. hispidula* and its close relative *G. subglabra.* The observed alterations regarding centromere and telomere sequences are in accordance with the presumed rapid evolution of genome size and organization as postulated previously by us (Vu *et al.*, 2015) and others (Ibarra-Laclette *et al.*, 2013).

RESULTS

A 161-bp tandem repeat and four CR elements were identified as putative centromeric sequences in *Genlisea nigrocaulis* and *G. hispidula*, respectively

Graph-based clustering analysis of unassembled randomly selected whole genome sequence reads revealed a different composition of repetitive DNA for the genomes of *G. nigrocaulis* and *G. hispidula* (Vu *et al.*, 2015). The *G. nigrocaulis* genome comprised only 15.9% repetitive DNA; 0.07% belong to the chromovirus clade of *Ty3/gypsy* long terminal repeat (LTR) retrotransposons, and 2.3% are

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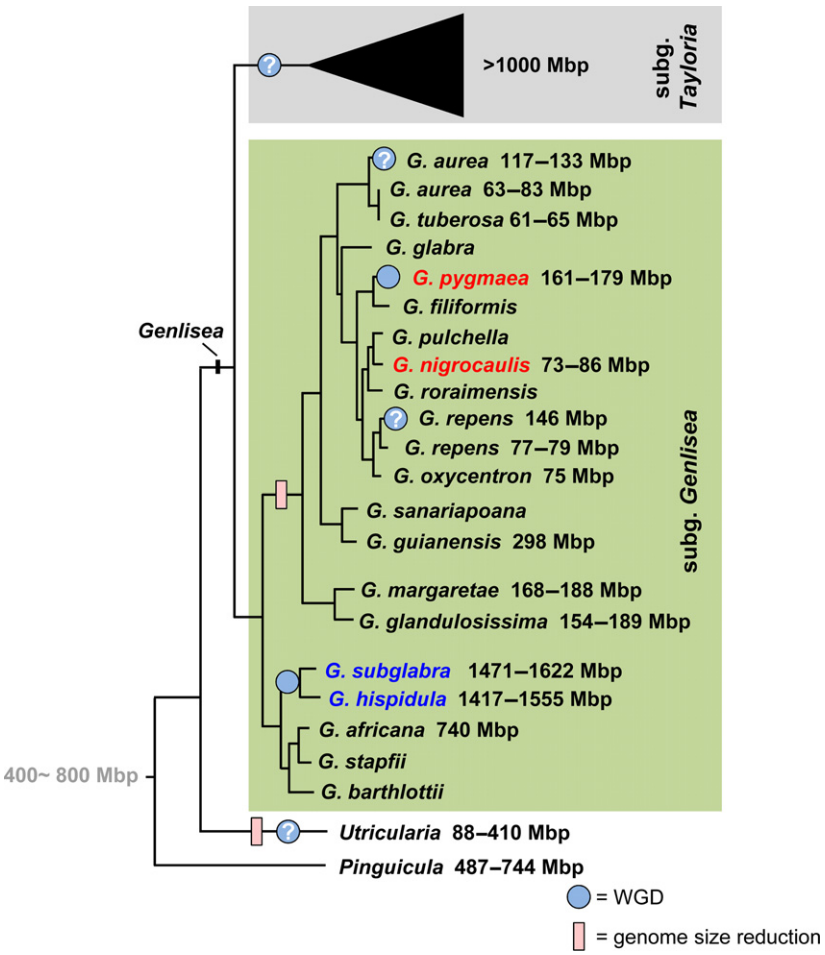


Figure 1. Simplified phylogenetic tree of the genus *Genlisea* modified according to Fleischmann *et al.* (2014) indicating switch points of genome diminution or confirmed (or presumed) whole genome duplication.
All numbers indicate genome sizes (Mbp) from Greilhuber *et al.* (2006), Fleischmann *et al.* (2014), Veleba *et al.* (2014), Tran *et al.* (2015), Vu *et al.* (2015) and own measurements.

tandem repeats. Among the few tandem repeats, a 161 bp tandem repeat (GnCent) is the most abundant one (Vu *et al.*, 2015). Used as FISH probe on *G. nigrocaulis* interphase nuclei, GnCent showed strong clustered signals colocalized with intensely 4′,6-diamidino-2-phenylindole (DAPI)-stained chromocenters (Figure 2a). In metaphase plates, GnCent yielded a distinct signal on every *G. nigrocaulis* chromosome (Figure 2b).

In *G. hispidula*, 64.1% of the genome were characterized as repetitive sequences, most of them as *Ty3/gypsy* LTR retrotransposons (24.64%) (Vu *et al.*, 2015). Among the only 1.96% of tandem repeats, several motifs were identified but none of them showed a hybridization signal on more than two chromosome pairs of *G. hispidula.* However, four families of *Ty3/gypsy* elements classified as members of the CRM clade (Gorinsek *et al.*, 2004) were found highly enriched in centromeric regions of *G. hispidula.* Further phylogenetic classification based on the reverse transcriptase domain of these retroelements showed that these elements, namely GhCR1, 2, 3 and 4, were clustered into a distinct branch of group A (Figure 3a), which was previously shown to include genuine CRs (Neumann *et al.*, 2011). Importantly, integrase C-termini of all four GhCR elements possessed a CR motif domain which is assumed to be responsible for targeting of centromeric retrotransposons to the centromeric region (Neumann *et al.*, 2011). This suggested that the four GhCR elements are constituents of the centromeres in *G. hispidula.* FISH analysis revealed clustered signals on each chromosome, mainly in a side-by-side arrangement, and some weaker dispersed signals of the individual or combined GhCR elements (Figure 3b,c).

The presence of GnCent and GhCR elements in other *Genlisea* species was tested by PCR and FISH analysis. The GnCent satellite was also amplifiable from genomic DNA (gDNA) of the tetraploid *G. pygmaea* (2n = 4x = 80, 179 Mbp/1C, Vu *et al.*, 2015), a close relative of *G. nigrocaulis*, but not from *G. aurea* and *G. margaretae*, all belonging to the lineage with small genomes (Figure 1). Except *G. nigrocaulis*, only chromosomes of *G. pygmaea* revealed FISH signals for GnCent (Figure 2f,g). Sequence comparison of GnCent repeats, cloned from PCR products amplified from gDNA of *G. nigrocaulis* and *G. pygmaea*,

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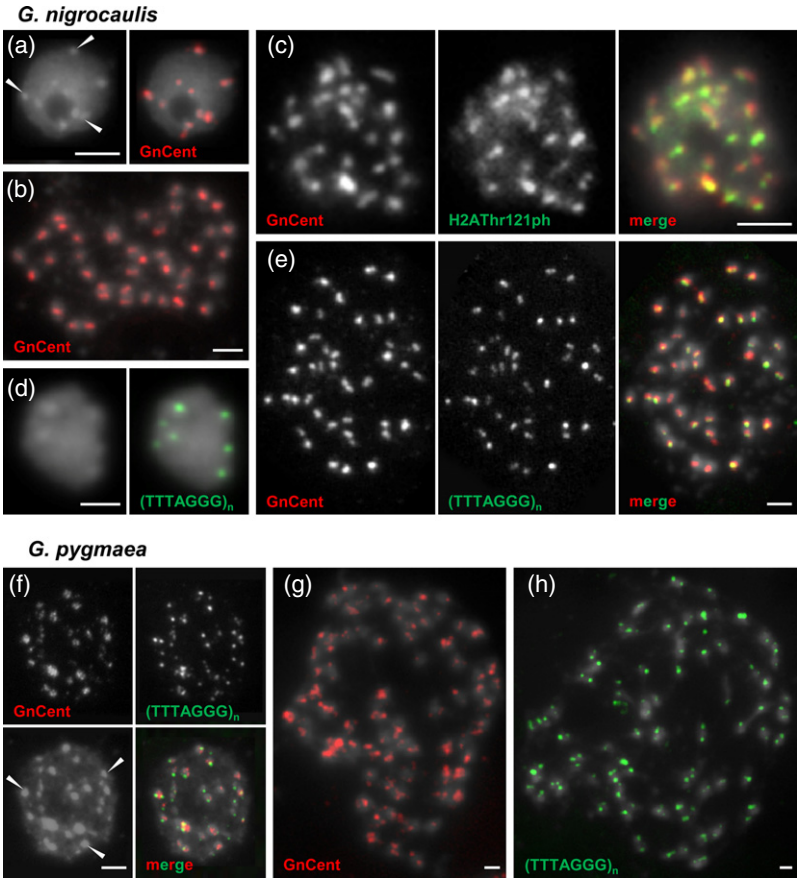


Figure 2. Cytogenetic characterization of centromeres and telomeres of *Genlisea nigrocaulis* and *G. pygmaea.*
(a, b) FISH with GnCent resulted in signal clusters co-localizing with intensely DAPI-stained chromocenters (white arrowheads) in interphase nuclei (a) and in a single hybridization signal per metaphase chromosome in *G. nigrocaulis* (b).
(c) Adjacent localization of GnCent FISH signals and H2AThr121ph immunostaining signals on metaphase chromosomes confirms the location of GnCent repeats in the (peri)centromere region of *G. nigrocaulis.*
(d, e) The Arabidopsis-type telomere repeat clusters in distinct foci in interphase nuclei (d) and localizes adjacent to the GnCent signals on metaphase chromosomes (e) of *G. nigrocaulis.*
(f) Also in *G. pygmaea* the GnCent signals co-localize with chromocenters and are closely associated with the Arabidopsis-type telomere repeat in interphase nuclei.
(g, h) These two repeats reveal strong FISH signals on each of the 80 metaphase chromosomes of *G. pygmaea.* Bars represent 2 μm.

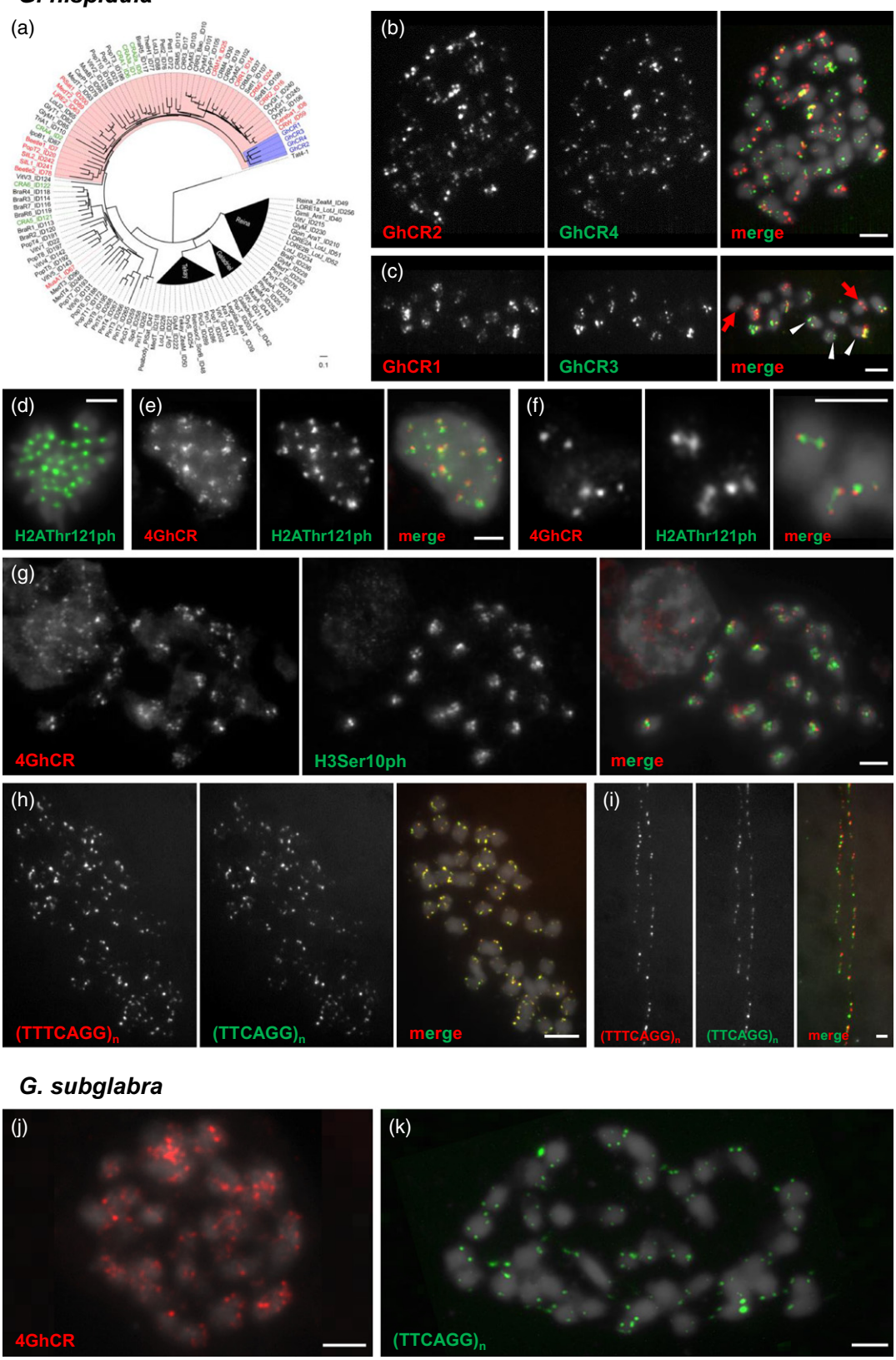
revealed an interspecific similarity of 85.2%. The variation between both species is mostly caused by a 3-bp insertion/deletion (Figure 4, blue box). Outside *G. hispidula*, the GhCR family was detectable by PCR and FISH only in *G. subglabra* (Figure 3j).

Pericentromere-specific histone markers localize adjacent to the putative centromere sequences of *Genlisea nigrocaulis* and *G. hispidula*

To confirm the presumed centromere specificity of the GnCent satellite in *G. nigrocaulis* and of the four GhCR elements in *G. hispidula*, we combined FISH of these repeats with immunostaining experiments using antibodies against phosphorylated threonine 121 of histone H2A (H2AThr121ph) and phosphorylated serine 10 of histone H3 (H3Ser10ph). Both modifications were previously shown to be highly conserved among plants and useful cytological markers to detect the (peri)centromeric region (Houben *et al.*, 1999; Demidov *et al.*, 2014). In *G. nigrocaulis*, we found adjacent signals for GnCent and H2AThr121ph on metaphase chromosomes (Figure 2c). Similarly, FISH signals of the four GhCR elements were immediately adjacent to immunosignals for H2AThr121ph on prophase nuclei and metaphase chromosomes

Figure 3. Cytogenetic characterization of centromeres and telomeres of *Genlisea hispidula* and *G. subglabra.*
(a) The neighbour-joining tree of the CRM clade, inferred from a comparison of reverse transcriptase domain sequences, reveals a distinct subgroup of GhCR family (blue shading) in group A (pink shading) comprising many elements with centromeric localization confirmed by FISH (red) or by *in silico* analysis (green) (modified from Neumann *et al.* (2011)).
(b) Combination of GhCR2 and GhCR4 shows signals clustered on each metaphase chromosome of *G. hispidula.*
(c) Partial metaphase of *G. hispidula* labeled by GhCR1 and GhCR3. Some chromosomes show enrichment of GhCR1 only (red arrows) or additional signals of either of the two sequences (white arrowheads).
(d–f) Anti-H2AThr121ph signals suggest monocentric chromosomes for *G. hispidula* (d) and localize together with adjacent FISH signals for the four GhCR elements (4GhCR) on prophase nuclei (e) and metaphase chromosomes (f).
(g) FISH signals of 4GhCR join those of H3Ser10ph, another (peri)centromeric-specific marker, in a partial metaphase of *G. hispidula.*
(h) Double-FISH detected the two non-canonical telomere sequence variants (TTCAGG, TTTCAGG) on *G. hispidula* chromosome ends.
(i) These variants are intermingled on extended chromatin fibers.
(j, k) Chromosomal distribution of 4GhCR signals (j) and the telomeric sequence variant TTCAGG in *G. subglabra* (k). Bars represent 3 μm.

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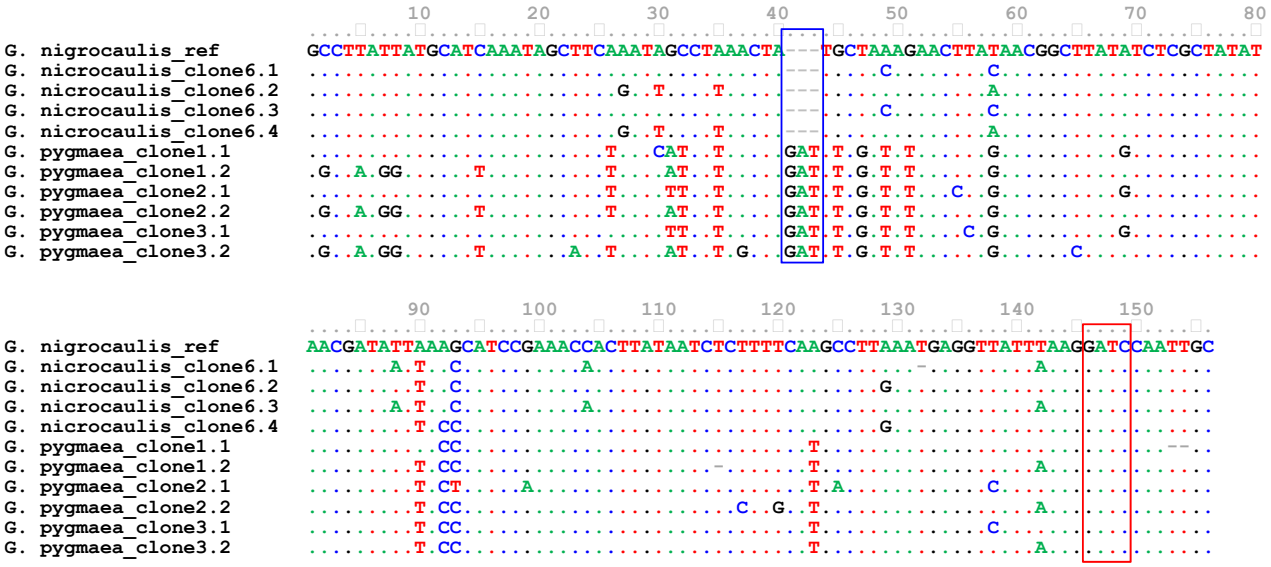


Figure 4. Sequence alignment of GnCent motifs shows significant variation between those cloned from gDNA of *Genlisea nigrocaulis* and *G. pygmaea*. The actual fragments amplified by a specific primer pair (Gn7c161, Table 1) were 153 bp and 156 bp for *G. nigrocaulis* and *G. pygmaea* clones, respectively. The blue and red boxes indicate the 3-bp insert/deletion region and the *Mbo*I recognition site, respectively.

(Figure 3d–f) and for H3Ser10ph on metaphase chromosomes (Figure 3g) in *G. hispidula*.

Genlisea nigrocaulis and *G. pygmaea* share the Arabidopsis-type telomere repeat

FISH using the Arabidopsis-type telomere sequence (TTTAGGG) as probe revealed strong signal clusters on interphase nuclei and on metaphase chromosomes of *G. nigrocaulis* (Figure 2d,e) and *G. pygmaea* (Figure 2f,h). On metaphase chromosomes of *G. nigrocaulis* the signals were found closely adjacent to the centromere-specific GnCent signals (Figure 2e).

Due to limitations in the availability of sufficient amount of leaf material of *G. nigrocaulis*, we used *G. pygmaea* to confirm the terminal position of the Arabidopsis-type telomere repeat by terminal restriction fragment (TRF) analysis. To detect TRFs and a possible association between telomere and the GnCent satellite sequence in *G. pygmaea*, digestion with the 6-bp cutter *Xba*I and the 8-bp cutter *Sfi*I (none of these have a recognition site in the GnCent satellite) was performed in BAL31-digested high-molecular-weight DNA. Using the Arabidopsis-type telomere sequence $(TAAACCC)_4$ as a probe, similar TRF patterns were found with both restriction enzymes (Figure 5a). Progressive TRF shortening with increasing BAL31 digestion times is visible in the pulse-field gel electrophoresis (PFGE) patterns (left panel) as well as in the blot from conventional electrophoresis (right panel); the latter detecting truncated TRFs diffusing out of the agarose plugs after BAL31 digestion. Re-probing of the membranes with GnCent revealed similar (but slightly stronger) hybridization patterns as for the telomeric probe (Figure 5c), thus supporting the close association of GnCent repeats and telomeric sequences on chromosomes. Since the restriction enzymes chosen for TRF analysis in *G. pygmaea* produce long fragments (comprising presumably tens of kbp of subtelomeric DNA), an additional restriction enzyme, the 4-bp cutter *Mbo*I, with a recognition site within the GnCent repeat (Figure 4, red box), was used to determine telomere lengths more precisely. TRFs ranging between 10 and 18 kbp were detected, reflecting the telomere lengths in this species (Figure 5b).

To demonstrate the presence of the Arabidopsis-type telomere sequence in *G. pygmaea* by a further independent approach, the TRAP (telomere repeat amplification protocol) assay was performed using telomerase extracts from either shoots with flower buds or young leafs, and from *A. thaliana* seedlings as a positive control (Figure 6a). Regular ladders of telomerase products were obtained in both extracts of *G. pygmaea*, showing the same periodicity of the ladder bands as the control products of *A. thaliana* telomerase (Figure 6a). The TRAP products were cloned and sequenced. The obtained sequences revealed regular $(TTTAGGG)_n$ repeats with four errors per 120 sequenced nucleotides (approximately 3% error rate).

Genlisea hispidula replaced the Arabidopsis-type telomeres by two sequence variants

Surprisingly, the Arabidopsis-type telomere sequence was not detectable by FISH, and among the genomic reads of *G. hispidula* only as short interstitial fragments. Instead, two variant repeat motifs of either 6-bp TTCAGG or 7-bp

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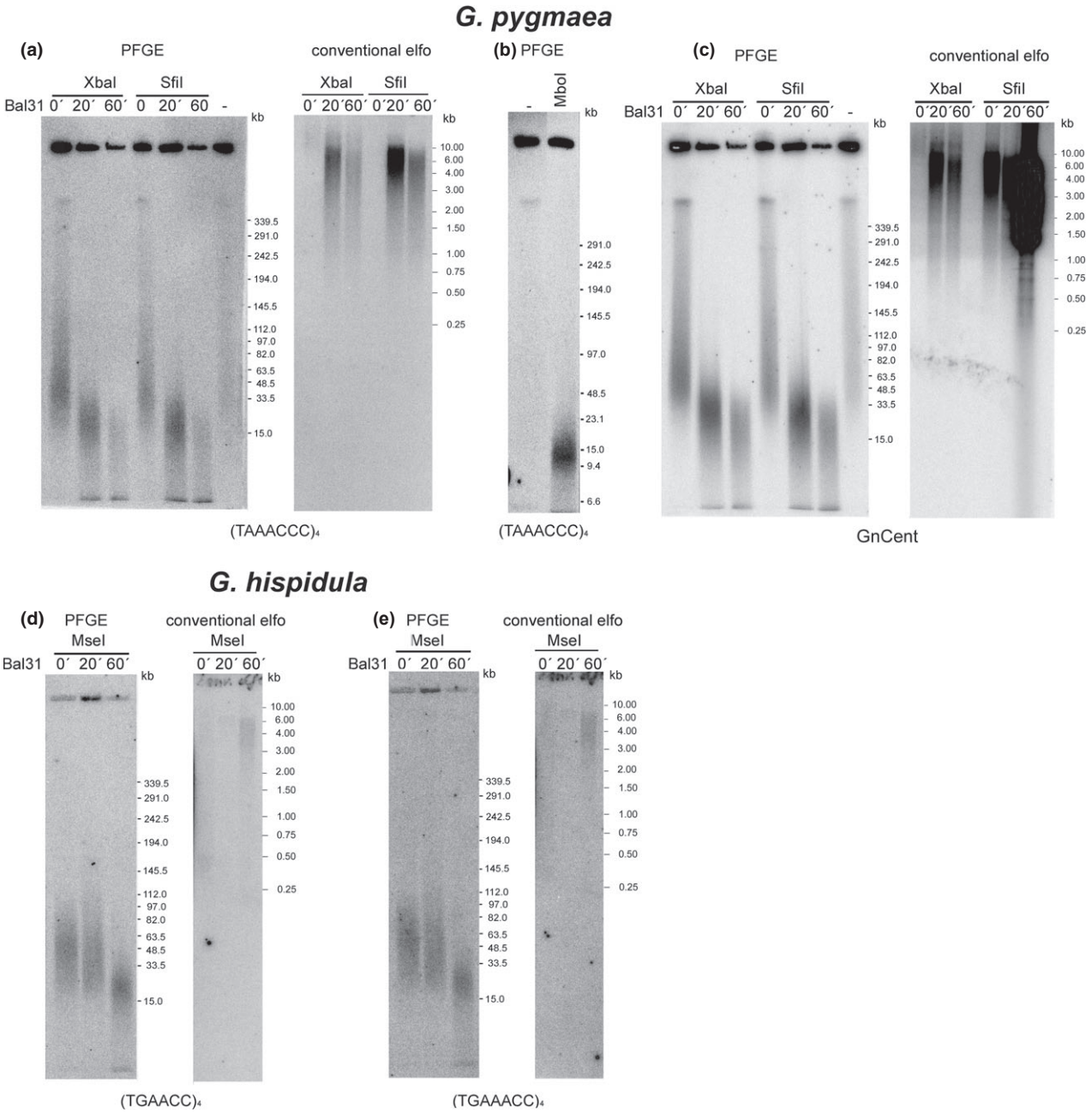


Figure 5. Telomere sequence analysis in *Genlisea pygmaea* and *G. hispidula.*

(a–c) BAL31 and TRF analysis in *G. pygmaea.* High-molecular-weight DNA in agarose plugs was digested with restriction enzymes *Xba*I or *Sfi*I without (–) or after prior BAL31 digestion for times indicated above individual lanes. PFGE denotes Southern hybridization of PFGE-separated DNA retained in agarose plugs after digestion, conventional electrophoresis denotes Southern hybridization of DNA diffused out of the plugs after digestion separated by conventional electrophoresis. Hybridization patterns obtained using the Arabidopsis-type telomere probe (a, b) and the putative centromeric repeat Gn7c161 (c) are shown (probed sequences are mentioned below each panel). Similar hybridization patterns of PFGE parts of panels (a) and (c) demonstrate association of both probed sequences in large TRF fragments, while TRF signal obtained using *Mbo*I (after cleavage of the 161-bp satellite) provides more precise assessment of genuine telomere lengths (between 10 and 18 kbp, b). Positions of LowRange PFG marker (NEB) and GeneRuler 1 kbp DNA ladder (Fermentas) are indicated on the right.

(d, e) BAL31 and TRF analysis in *G. hispidula.* High-molecular-weight DNA in agarose plugs was digested with *Mse*I restriction enzyme without (–) or after prior BAL31 digestion for times indicated above individual lanes. PFGE denotes Southern hybridization of PFGE-separated DNA retained in agarose plugs after digestion; conventional electrophoresis denotes Southern hybridization of DNA diffused out of the plugs after digestion, separated by conventional electrophoresis. Hybridization patterns obtained using the short (d) and the long (e) variants of telomeric repeat units are shown (probed sequences are mentioned below each panel). Positions of MidRange PFG marker (NEB) and GeneRuler 1 kbp DNA ladder (Fermentas) are indicated on the right.

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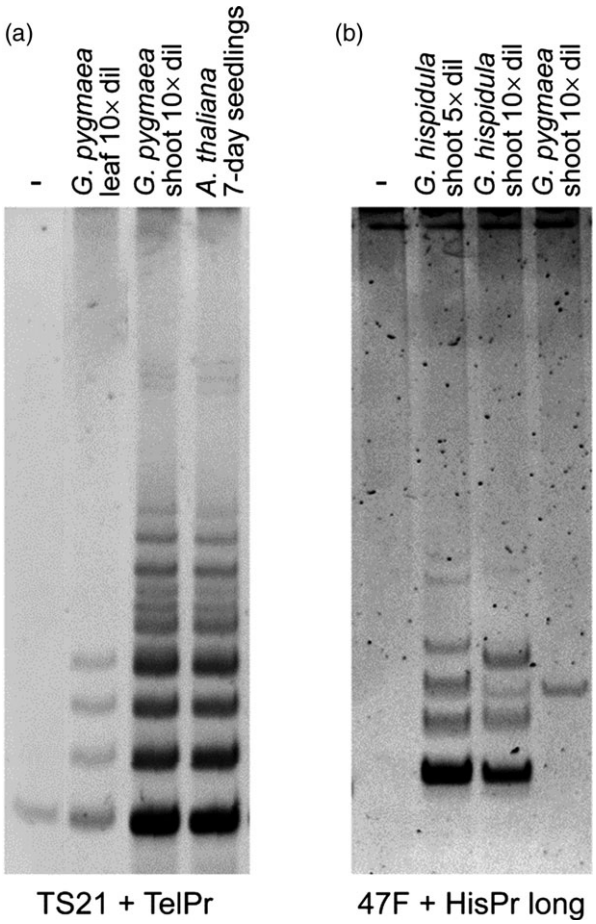


Figure 6. Telomerase activity detection in *Genlisea pygmaea* (a) and *G. hispidula* (b) using TRAP assay.
Substrate and reverse primers used in TRAP are given below each panel (see Table 1). *G. pygmaea* shows regular ladders of telomerase products, similar to *A. thaliana* extract used as positive control (a). While *G. hispidula* extracts produce irregular ladders of telomerase products, *G. pygmaea* extract shows only a single band of a non-specific product with *G. hispidula* primers (b).

TTTCAGG could be identified as putative telomere sequences. FISH using these motifs as probes indicated their terminal location on metaphase chromosomes of *G. hispidula* and of its close relative *G. subglabra* (Figure 3h,k). Fiber FISH experiments demonstrated that both sequences are intermingled (Figure 3i). Probes of both motifs, $(TGAAACC)_4$ and $(TGAACC)_4$, revealed similar patterns of TRFs ranging from 20 to 97 kbp at individual chromosome ends. Their sensitivity to BAL31 digestion proved the terminal chromosome position of the hybridizing fragments (Figure 5d,e). During the longest BAL31 exposure, the truncated TRFs diffuse from agarose plugs and can be detected in the DNA fraction obtained from the reaction solution (Figure 5d,e), right panels). No signal could be observed when re-probing the membranes with the Arabidopsis-type telomere sequence $(TAAACCC)_4$.

The identity of telomeric sequences was further tested using the TRAP assay with *G. hispidula* telomerase extracts and subsequent sequencing of the resulting products. Several combinations of substrate and reverse primers have been tested, but positive results were only obtained when the primer combinations 47F as substrate and either HisPr long or HisPr short as reverse primer (see Table 1) were used. Finally, the primer for the longer repeat variant, HisPr long, yielding well reproducible results, was chosen for the TRAP assay (Figure 6b). No product was obtained in the reaction without protein extract, and in the extract from *G. pygmaea* only a single, most likely unspecific band was detected (Figure 6b). *G. hispidula* displayed irregular ladders, presumably due to alternation of long and short variants of the synthesized telomere repeat (Figure 6b). Sequencing of the products demonstrated co-occurrence of both variants in all clones that contained more than one sequenced repeat unit. While it is not clear whether the observed low processivity of *G. hispidula* telomerase is due to *in vitro* conditions, or is an intrinsic feature of this telomerase, the observed alternation of both telomeric sequence variants confirmed next-generation sequencing results (see Table S2), as well as the co-localization of both sequences observed in FISH and TRF analyses.

DISCUSSION

Two small *Genlisea* genomes revealed predominantly tandem repeats, and two large genomes clusters of retroelements at their centromeres

Centromere-specific repeats were not found on *Tradescantia* chromosomes (Houben *et al.*, 1996). *De novo* centromere formation in barley may occur at positions that do not contain the centromeric repeats typical for this species (Nasuda *et al.*, 2005). Non-repetitive DNA sequences were characterized as centromere-specific for five chromosomes of potato (Gong *et al.*, 2012). Sequence and abundance of centromeric repeats can vary, even between centromeres of the same species (Berr *et al.*, 2006; Neumann *et al.*, 2012). Nevertheless, centromeric repeats and/or CR elements are the major constituents of plant centromeres, although their contribution can vary between chromosomes of a complement (Plohl *et al.*, 2014). In *G. nigrocaulis*, and in its tetraploid relative *G. pygmaea*, the 161 bp GnCent satellite apparently coincides with the centromere position. Taking into account the rapid divergence of centromeric satellite repeats (Henikoff *et al.*, 2001), the presence of the 161 bp tandem repeat in both species suggests their close relationship with a common ancestor. In *G. hispidula* FISH revealed GhCR elements in positional coincidence with immunosignals for the (peri)centromere marks H2AThr121ph and H3Ser10ph. The presence of these GhCRs on each chromosome of the close relative

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Table 1 Sequences of primers used to generate FISH probes (a) or in analyses of telomerase activity in tissues of *Genlisea hispidula* and *G. pygmaea* (b)

(a) Primers for FISH probes

Name	Sequence (5′→3′)	Product
Gn7c161	F: GCCTTATTATGCATCAAATAGCTTC; R: GCAATTGGATCCTTTAATAACCTC	161 bp tandem repeat (GnCent)
Gh_CR01	F: GCTTTGACAGAACCCCTTAATC; R: AGCTTCCTTGGTCTCCACTT	Centromeric retrotransposon element (GhCR1, 4.8 kbp)
Gh_CR02	F: TGTCCTCAAGTAAAGATAAAGAAAAA; R: TCCACCTCAAATGAGTTCATAA	Centromeric retrotransposon element (GhCR2, 4.8 kbp)
Gh_CR03	F: TCTAGATAGCAAGATGATCCTTGAG; R: TTAGGCTCCTCGAATGTGAT	Centromeric retrotransposon element (GhCR3, 4.8 kbp)
Gh_CR04	F: ATGTCCAAGGATAAGGTGATTG; R: CTTGAGTGGTTCTTGCTTGAT	Centromeric retrotransposon element (GhCR4, 5.2 kbp)
Atha_TEL	F: TTTAGGGTTTAGGGTTTAGGGTTTAGGG; R: CCCTAAACCCTAAACCCTAAACCCTAAA	Arabidopsis-type telomere
Gh_TEL6	F: TTCAGGTTCAGGTTCAGGTTCAGG; R: CCTGAACCTGAACCTGAACCTGAA	6 bp telomeric variant of *G. hispidula*
Gh_TEL7	F: TTTCAGGTTTCAGGTTTCAGGTTTCAGG; R: CCTGAAACCTGAAACCTGAAACCTGAAA	7 bp telomeric variant of *G. hispidula*

(b) Primers for analyses of telomerase activity

Name	Sequence (5′→3′)	Species
TS21	S: GACAATCCGTCGAGCAGAGTT	*G. pygmaea*
TelPr	R: CCGAATTCAACCCTAAACCCTAAACCCTAAACCC	*G. pygmaea*
47F	S: CGCGGTAGTGATGTGGTTGTGTT	*G. hispidula*
HisPr long	R: CCGAATTCAAACCTGAAACCTGAAACCTGAAACC	*G. hispidula*
HisPr short	R: CCGAATTCTGAACCTGAACCTGAACCTGAACC	*G. hispidula*

F: forward primer; R: reverse primer; S: substrate primer.

G. subglabra suggests that the last common ancestor of both species already recruited these elements as centromeric sequence.

While retrotransposons became accumulated at centromere positions in the *G. hispidula* clade, they were likely eliminated in the course of genome shrinking within the *G. nigrocaulis* clade during evolution from an ancestor with an intermediate genome size of 400–800 Mbp (Vu *et al.*, 2015). Additional weak and dispersed FISH signals of GhCRs along several metaphase chromosomes of *G. hispidula* might indicate remnants of the ancestral state or cross-hybridization with related elements (Figure 3c).

Two large *Genlisea* genomes display novel tandem repeats at their chromosome ends

In contrast to centromeric DNA, telomeric sequences are usually composed of highly conserved simple sequence repeats and vary little between different eukaryotic kingdoms. For instance, the vertebrate telomere sequence $(TTAGGG)_n$ differs only by one T from the motif $(TTTAGGG)_n$ of *Arabidopsis* (Richards and Ausubel, 1988) which is typical for most of the investigated plant species (Fuchs *et al.*, 1995). First evidence that not all plant species protect their chromosome ends by the Arabidopsis-type telomere came from FISH with a $(TTTAGGG)_n$ probe on chromosomes of several species of the genus *Allium* (Fuchs *et al.*, 1995). Pich *et al.* (1996) and Pich and Schubert (1998) confirmed the absence of $(TTTAGGG)_n$ and suggested that highly repetitive satellite and/or rDNA sequences substituted the original telomeric sequence at the very ends of the *Allium* chromosomes. They also suggested a conversion-like mechanism for compensation of replication-mediated end-shortening. Later, Adams *et al.* (2000) reported the lack of Arabidopsis-type telomeres in *Aloe*, another genus of the order Asparagales to which *Allium* belongs. A large-scale screen using different minisatellite sequences identified the vertebrate-type telomere as the predominant satellite repeat, occasionally associated with (TTTAGGG) repeats, at the chromosome termini of Asparagales species which descend from an evolutionary switch-point leading to an extensive replacement of the Arabidopsis motif (Sykorova *et al.*, 2006). A second switch-point (inside the Alliaceae family) towards the evolution of the genus *Allium* is responsible for the loss of any of the known telomere minisatellites, and apparently also of an active telomerase gene in this genus (Fajkus

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et al., 2005). Also three genera of the Solanaceae family, *Vestia*, *Sessea* and *Cestrum*, lack Arabidopsis- and vertebrate-type telomeres (Sykorova *et al.*, 2003a,b). Recently, an A/T rich (TTTTTTAGGG) minisatellite was characterized as telomeric sequence for *Cestrum elegans* (Peska *et al.*, 2015). However, it still needs to be confirmed whether this non-canonical telomeric repeat is also shared by the two sister genera *Vestia* and *Sessea.*

In the genus *Genlisea* we found in *G. nigrocaulis*, and in the closely related *G. pygmaea*, Arabidopsis-type telomere repeats. In the much larger genome of *G. hispidula*, and in that of its close relative *G. subglabra*, this sequence motif is replaced at the chromosome ends by two hitherto unknown telomeric sequence variants. In Asparagales, the change from the Arabidopsis-type (TTTAGGG) to the vertebrate-type (TTAGGG) telomere can be explained by point mutations in the telomerase catalytic protein subunit (TERT) which affect the telomere-telomerase interaction and result in a variant usage of template region in the telomerase RNA subunit (TR) and consequently in T-slippage (Sykorova *et al.*, 2006). In *Genlisea* however, the variant motifs observed in *G. hispidula* cannot be derived from the presumed ancestral Arabidopsis-type telomere sequence by less than two mutations. In particular, the presence of cytosine in the G-rich strand of both sequence variants is remarkable. This change is most likely associated with a corresponding mutation directly in the template region of the *G. hispidula* TR subunit. Variants having a C within the G-rich strand are rare (http://telomerase.asu.edu/sequences_telomere.html), and were to our knowledge hitherto not described for plants. An evolutionary switch of telomeric sequences within a genus has also not been reported so far (Figure 7). This switch occurs in parallel with an apparently high mutation rate (Ibarra-Laclette *et al.*, 2013) and a fast genome size evolution within the genus *Genlisea* (Vu *et al.*, 2015). Telomeric sequences of the neighbor clade (*G. africana*, *G. stapfii* and *G. barthlottii*) could not be investigated due to the lack of available plant material. Therefore, it remains an open question whether the new telomere repeat variants evolved immediately before separation of *G. hispidula* and *G. subglabra* or already earlier, at the basis of that branch of the subgenus *Genlisea* that displays large genomes (Figure 1).

Conclusions

The presented data strongly argue that centromeres and telomeres are evolutionarily flexible in a hitherto unknown scale, with different sequences being recruited to perform the same function. Whereas in a group of species of the genus *Genlisea* with very small genomes (86–179 Mbp) a tandem repeat occupies the centromeric position, in species of a group with almost 20-fold larger genomes four retroelements were found in the centromeres instead. Thus, even within the same genus, either tandem repeats

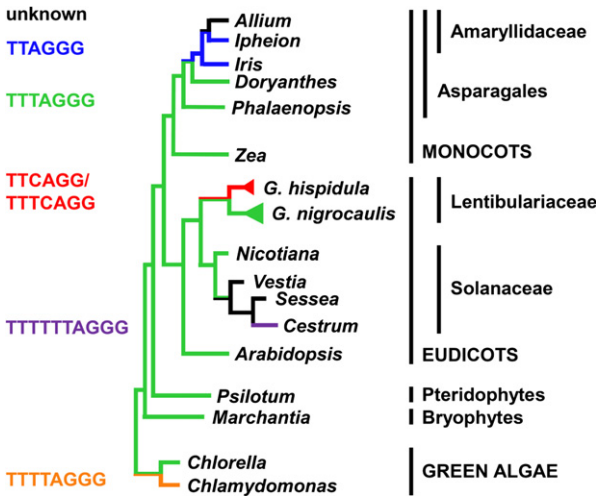


Figure 7. Phylogenetic tree of green plants indicating telomeric sequence deviations from TTTAGGG, the most widespread telomeric repeat among plants [modified according to Sykorova *et al.* (2006) and Peska *et al.* (2015)].

or retroelements can serve as centromeric sequences. Surprisingly, both groups differ also in the sequence of their telomeric repeats. The canonical plant telomeric repeat TTTAGGG, present at the chromosome termini of the small genomes, is replaced by two intermingled sequence variants (TTCAGG and TTTCAGG) in species with the large genomes. Hence, also between closely related species intrageneric switches of telomeric repeats may occur. The presence of cytosine within the guanine-rich strand of the two described sequence variants has potential implications for the modification of telomeric chromatin.

EXPERIMENTAL PROCEDURES

Plant material

Plants of species used in this study (*G. nigrocaulis*, *G. hispidula*, *G. pygmaea*, *G. subglabra)* were obtained from Best Carnivorous Plants Store (http://www.bestcarnivorousplants.net; Ostrava, Czech Republic): *G. nigrocaulis*, *G. hispidula*, *G. pygmaea;* Carnivores and more (http://www.carnivorsandmore.de; Merzig, Germany): *G. nigrocaulis*, *G. subglabra* and Die Welt der Fleischfressenden Pflanzen (http://www.falle.de; Gartenbau Thomas Carow, Nüdlingen, Germany): *G. nigrocaulis.* All species were cultivated in a greenhouse. Herbarium vouchers of *G. nigrocaulis*, *G. pygamaea* and *G. hispidula* were deposited at the IPK Gatersleben (Table S1).

Preparation of genomic DNA, nuclei and chromosomes

Genomic DNA of *Genlisea* species was isolated using the DNeasy® Plant Mini kit (Qiagen, https://www.qiagen.com/). Concentration and quality of the DNA were estimated using a NanoDrop spectrophotometer (Thermo Scientific, http://www.thermoscientific.com/) and by 1% (w/v) agarose-gel electrophoresis.

Interphase nuclei from leaves were isolated after formaldehyde fixation, flow sorted and dropped onto slides as described by (Lysak *et al.*, 2006). Slides were stored at −20°C until use.

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For chromosome preparations either flower buds or very young leaf tips were used. Squashing was performed as described in Lysak *et al.* (2006) with minor modifications. In brief, plant material was treated in 20 mM 8-hydroxylquinonline for 2 h at room temperature followed by 4 h at 4°C to accumulate dividing cells. After washing briefly in distilled water, the material was fixed in ethanol: glacial acetic acid (3:1) for at least 24 h at room temperature. Fixed tissues were used immediately or stored at 4°C for several days. After three washes in citrate buffer (10 mM sodium citrate, pH 4.5) for 5 min each, the plant tissue was softened in 2% pectinase and 2% cellulose, (w/v) in citrate buffer, at 37°C, for 10 min (young leaves) or 15 min (flower buds). The digestion was stopped by adding ice-cold citrate buffer. Squashing on microscopic slides was performed in a drop of 75% glacial acetic acid. After freezing in liquid nitrogen, slides were rinsed in 2 × SSC, dehydrated in an ethanol series (70, 90 and 96%) and air dried. The quality of spreading was evaluated after DAPI staining. Slides harboring more than 10 well-spread metaphases were stored at 4°C until use for FISH. For immunofluorescence experiments, chromosomes were prepared according to Houben *et al.* (2003). For preparation of extended chromatin fibers, suspensions of nuclei were prepared according to Lysak *et al.* (2006), dropped on one end of a microscopic slide and dragged along the slide as described (Li *et al.*, 2005).

Fluorescence *in situ* hybridization (FISH) and immunostaining

To generate FISH probes, the GnCent repeat and four GhCR elements identified by graph-based clustering analysis (Novak *et al.*, 2010; Vu *et al.*, 2015) were PCR amplified with sequence-specific primers (Table 1) using either a GoTag Kit (Promega) or a Phusion High Fidelity DNA Polymerase Kit (Thermo Scientific). PCR products were analyzed by 1% (w/v) agarose-gel electrophoresis, cloned by pGEM-T Easy Vector Systems (Promega, http://www.promega.com/) and confirmed by Sanger sequencing. Telomere probes were amplified using tetramers of the telomeric sequences in the absence of DNA template as described by Ijdo *et al.* (1991). Probes were labeled by nick-translation (Lysak *et al.*, 2006) using either TexasRed-dUTP, Alexa 488-dUTP (Life Technologies, http://www.thermofisher.com/) or Cy3-dUTP (Amersham, http://www.gelifesciences.com/). FISH was performed according to Lysak *et al.* (2006).

For immunostaining, slides were incubated in blocking buffer (5% horse serum, 3% BSA in 1 × PBS) immediately after preparation for 1 h at 37°C, shortly washed in 1 × PBS and incubated with primary antibodies for 16 h at 4°C in a humid chamber. As primary antibodies rabbit anti-H2AThr121ph (MyBioSource, http://www.mybiosource.com/, cat-No. MBS004447, dilution 1:1000, considering the removal of the initial N-terminal methionine it corresponds to H2AThr120ph) and rabbit anti-H3Ser10ph (Merck Millipore, http://www.merckmillipore.com/, cat-No. 06-570, dilution 1:50) were used. Subsequently, slides were washed in 1 × PBS three times for 10 min each and then incubated with Alexa 488-conjugated anti-rabbit antibody (Life Technologies, dilution 1:200) in a humid chamber at 37°C for 1 h. After final washes in 1 × PBS (three times, 10 min each), slides were counter-stained with DAPI (1 μg ml^{-1}) in VectaShield.

Microscopy and image processing

FISH and immunostaining preparations were analyzed using a Zeiss Axioplan 2 epifluorescence microscope equipped with a cooled CCD camera (Diagnostic Instruments, Inc., http://www.spotimaging.com/). Fluorescence images for each fluorochrome were captured separately using appropriate filter combinations and a 100× objective. The images were processed (brightness and contrast adjustment only), pseudo-colored and merged using ADOBE PHOTOSHOP software (Adobe Systems, http://www.adobe.com/).

Analysis of telomere lengths and BAL31-sensitivity

High-molecular-weight DNA from *Genlisea* plants was prepared in agarose plugs as described (Sykorova *et al.*, 2006). Briefly, plant tissues were homogenized in liquid nitrogen, resuspended in homogenization buffer [10 mM 2(*N*-morpholino) ethanesulphonic acid (MES), pH 5.6; 10 mM NaCl; 5 mM EDTA], mixed with an equal volume of 2% (w/v) low melting point agarose in 400 mM mannitol, 20 mM MES, pH 5.6 and transferred to a mould. Samples in agarose blocks were treated by proteinase K (0.1 mg ml^{-1}) in TES buffer [0.5 M EDTA, pH 8.0; 10 mM Tris–HCl, pH 8.0; 1.0% (w/v) lauroylsarcosine] at 37°C for 24 h. Deproteinised plugs were washed twice in TE buffer (10 mM Tris-HCl, pH 8.0; 1 mM EDTA, pH 8.0) for 30 min, twice in TE with 1 mM phenylmethylsulfonyl fluoride for 30 min and 3 × 30 min in 0.1 × TE. The terminal position of probed sequences was tested by BAL31 nuclease sensitivity. DNA samples in agarose plugs (approximately 2 μg) were equilibrated for 15 min in 200 μl of BAL31 nuclease buffer (NEB, cat-No. M0213S) and then digested in a new portion of buffer with 3 units of BAL31 nuclease for 20 or 60 min in a thermomixer (Eppendorf, http://www.eppendorf.com/) at 30°C. The reactions were stopped by buffer exchange with 50 mM EGTA pH 8.0, and BAL31 was irreversibly inactivated by incubation at 58°C for 30 min. Then the plugs were washed in 0.1 × TE buffer (three times for 15 min) and equilibrated in appropriate restriction buffer for subsequent restriction enzyme digestion to measure telomere lengths by TRF analysis (Fajkus *et al.*, 1998). After digestion, the low-molecular-mass fraction of digested DNA was precipitated with ethanol and dissolved in TE for analysis by conventional agarose-gel electrophoresis and Southern hybridization. High-molecular-mass fractions were retained in the agarose plugs, and analyzed by PFGE using the BioRad Chef Mapper instrument under the following conditions: 1% (w/v) FastLane agarose gel in 0.5 × TBE buffer, 6 V/cm, pulse time ramped from 0.5 to 26 sec at 14°C for 20 h. Conventional and PFGE gels were alkali blotted and hybridized with radioisotope-labeled probes (Fojtova *et al.*, 2010). Hybridization signals were visualized with a FLA-7000 phosphofluoroimaging system (Fuji Film, http://www.fujifilm.com/).

Telomere repeat amplification protocol

Leaves and shoots of *Genlisea* species were manually homogenized in extraction buffer according to Fitzgerald *et al.* (1996). Crude extracts obtained after centrifugation were 5× and 10× diluted for analysis of telomerase activity. Primers usually applied in Telomere repeat amplification protocol (TRAP) assays for plants (Fajkus *et al.*, 1998) were used for analysis of *G. pygmaea.* Additional primers and primer combinations corresponding to putative telomere repeats in *G. hispidula* were tested (Table 1). One μl of 10 μM substrate primer (TS21 for *G. pygmaea* and 47F (Fojtova *et al.*, 2002) for *G. hispidula*) was mixed with 1 μl of diluted crude protein extract and elongation of the primer by the telomerase proceeded for 45 min at 26°C in 25 μl reaction buffer as described (Fojtova *et al.*, 2010). After heat inactivation of telomerase (5 min, 94°C), 1 μl of 10 μM reverse primer (TelPr for *G. pygmaea;* HisPr short or HisPr long for *G. hispidula*) and 2U of DyNAzymeII DNA polymerase (Finnzymes, http://www.thermofisher.com/) were added and extension products were amplified (35 cycles of 95°C/30 sec, 65°C/30 sec, 72°C/30 sec; final extension 72°C/5 min). Aliquot samples of TRAP reactions were analyzed on 12.5% (w/v)

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polyacrylamide gel in 0.5 × TBE buffer. Gels were stained by GelStar Nucleic Acid Gel Stain (LONZA) and signals were visualized using the LAS-3000 system (FujiFilm).

Products of TRAP were cloned using a TOPO-TA cloning kit (Invitrogen, https://www.thermofisher.com/) and sequenced to characterize sequences added by telomerase.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interests.

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Table S1. List of herbarium vouchers of *Genlisea* species deposited at the IPK Gatersleben.

Table S2. Analysis of reads with telomeric motifs GGTTTCA and GGTTCA in *Genlisea hispidula.*

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